

6 System description: Payroll Management System

At PayThemAll, Inc., the workday begins differently than it once did. The company's old payroll platform had become unreliable, so the IT department was tasked with building something entirely new. Today, when employees arrive at work, they take out their phones or tablets and log into the system using their company email and password. For additional security, the login requires two-step verification—either through an SMS code or a biometric scan on supported devices. The screen that greets them is familiar yet modern: their current timecard is displayed, waiting to be filled. If it happens to be a new pay period, the system has already created the timecard for them, with the start and end dates locked in place. Employees can add hours or project information as the week goes on, but once the card is submitted, it cannot be touched again—it becomes read-only, protecting both the company and the worker from disputes.

Hourly workers, in particular, depend on this. Ali, who works in operations, enters his daily hours carefully. He knows the system will not let him enter more than seven hours for a single day, but he also knows that any overtime beyond that threshold will earn him twice his usual pay. The system automatically sends him a weekly summary notification, showing his hours, overtime, and projected earnings before payroll runs. Every Saturday, like clockwork, the system processes his week's timecard and ensures that his wages are deposited into his bank account by the following Monday morning. If there is any error—such as a missing entry—Ali is alerted immediately so he can correct it before processing.

Salaried employees interact with the system differently. Saleh, a project manager, receives the same paycheck on the second working day of each month, but he still fills out a timecard. His hours are not used to calculate pay but to track how time is spent across the company's many projects. This information is particularly valuable since the payroll system is connected to the company's legacy Work Management Database, which contains all the project charge numbers. Although the payroll application may read from this database, it is never allowed to alter its data. Saleh can also export his hours into project dashboards, where his department tracks progress and budgets in real time.

For those working on commission, the story unfolds differently again. Fahad, a sales associate, uses the system's purchase order feature. Each time he makes a sale, he records the date, the product, the amount, and the customer's billing address. The system assigns his order a unique number—a reference he can use later to adjust or even delete the record if needed. A built-in dashboard helps him track which orders are still pending approval, which have been processed, and how much commission he has earned so far in the month. When payday arrives on the second working day of the month, the system calculates his commission rate—whether 10, 15, 25, or 35 percent—and adds it to his base pay.

Beyond recording time and sales, employees appreciate the system's reporting capabilities. Ali, curious about how many hours he has logged this year, requests a report. He specifies the type—"total hours worked"—along with a date range. Within moments, the system generates a clean report that he can save, print, or export to Excel. Other employees often use this feature to check year-to-date pay, vacation balances, remaining sick

days, or hours devoted to a particular project. To avoid errors, the system includes built-in validation: if someone requests a report over an unsupported time range, it suggests the correct options automatically.

While employees control their own records, some responsibilities remain in the hands of the payroll administrator. After logging in, Raed, the administrator, can add a new hire, remove an employee who has left, or update records such as salary, commission rate, hourly pay, or even tax deductions. He can also manage employee benefits, such as health insurance enrollment and retirement plan contributions, directly through the payroll interface. Beyond administrative updates, he can generate high-level reports, for example, to review the total hours worked by a specific team, to confirm year-to-date payroll figures, or to analyze trends in overtime across departments. For sensitive actions, the system requires Raed to re-authenticate using his administrator token, ensuring that no unauthorized changes are made. Employees cannot see or modify such details; their control is limited to their own timecards and their chosen payment method, whether direct deposit or collecting a check at the office.

Every Saturday morning and again on the second working day of the month, the system quietly carries out its most critical function: running payroll. Drawing upon submitted timecards, purchase orders, and employee data, it calculates each paycheck, applies deductions, and prepares payments. Integration with external banking systems ensures that deposits are transferred electronically without the need for manual intervention. Reliability is paramount—if the system were to fail during payroll hours, thousands of employees would be affected. For this reason, it has been designed to support simultaneous access across the company, remain available during processing times, and automatically create backup copies of payroll data in secure cloud storage.

Through this new system, PayThemAll has turned what was once a cumbersome process into an efficient, automated workflow. Employees of every type—hourly, salaried, and commissioned—now have the tools they need at their fingertips, while Raed and his team retain oversight and control. With its layered security, mobile compatibility, smart notifications, and cloud resilience, the platform feels less like an old back-office system and more like a reliable daily companion in the rhythm of work.

Glossary

The glossary contains the working definitions for the key concepts in the PayContract Payroll Management System:

- **Bank System:** Any bank to which direct deposit transactions are sent.
- **Employee:** A person who works for the company that owns and operates the payroll system (PayContract, Inc.)
- **Payroll Administrator:** The person responsible for maintaining employees and employee information in the system.

- **Work Management Database:** The legacy database that contains all information regarding projects and charge numbers.
- **System Clock:** The internal system clock that keeps track of time. The internal clock will automatically run the payroll at the appropriate times.
- **Pay Period:** The amount of time over which an employee is paid.
- **Paycheck:** A record of how much an employee was paid during a specified Pay Period.
- **Payment Method:** How the employee is paid, either pick-up, or direct deposit.
- **Timecard:** A record of hours worked by the employee during a specified pay period.
- **Purchase Order:** A record of a sale made by an employee.
- **Salaried Employee:** An employee that receives a salary.
- **Commissioned Employee:** An employee that receives a salary plus commissions.
- **Hourly Employee:** An employee that is paid by the hour.